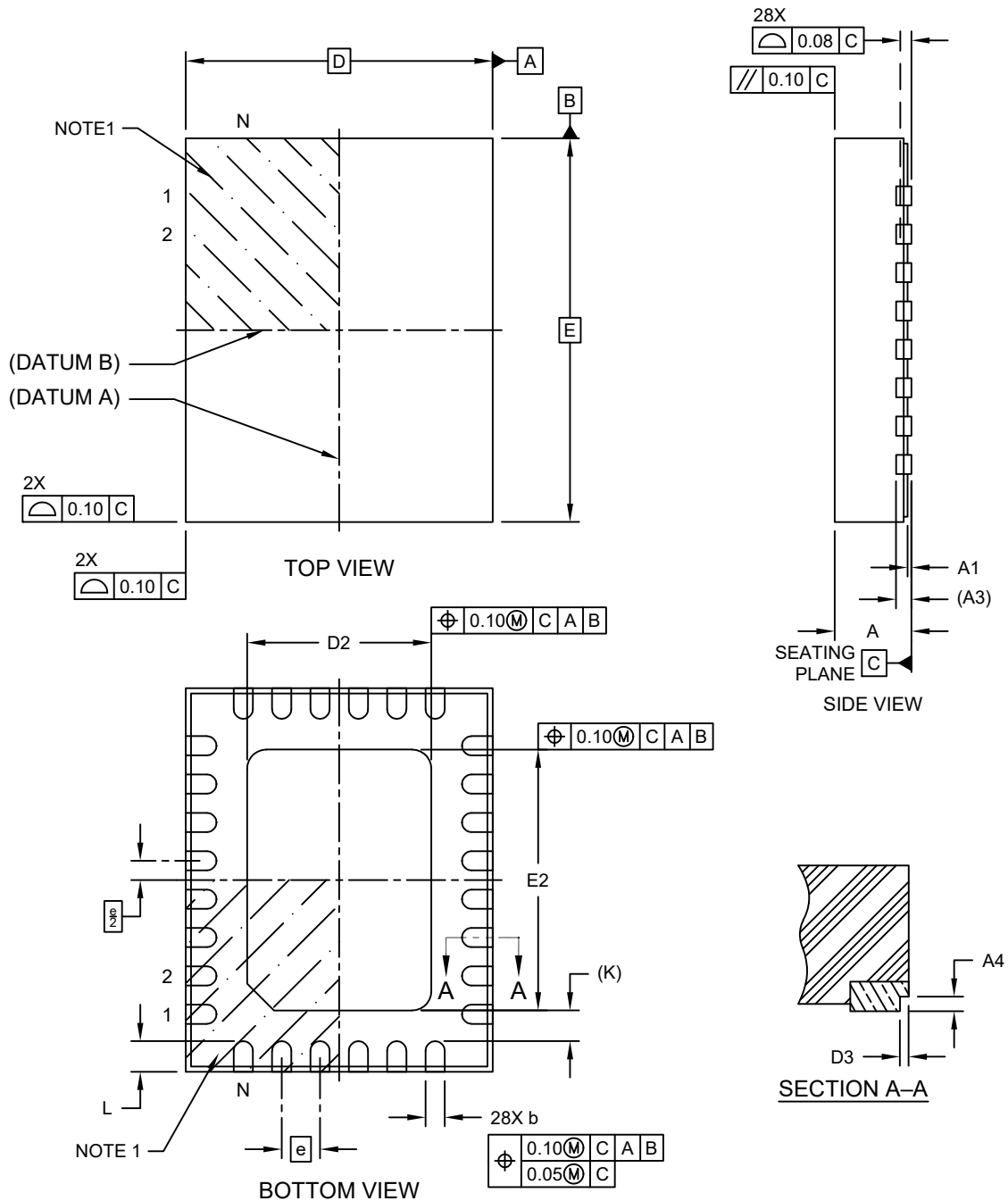


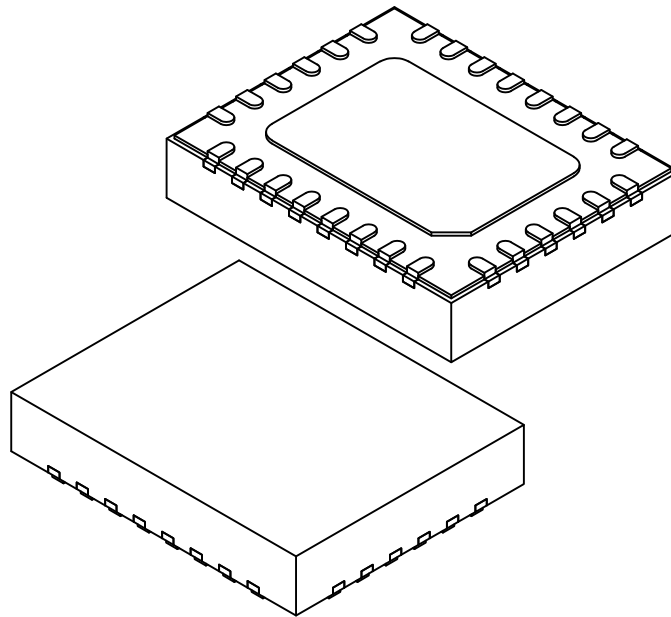
# **28-Lead Very Thin Plastic Quad Flat, No Lead Package (HPB) - 4x5x1.0 mm Body [VQFN]** **With 2.4x3.4 mm Exposed Pad and Stepped Wettable Flanks**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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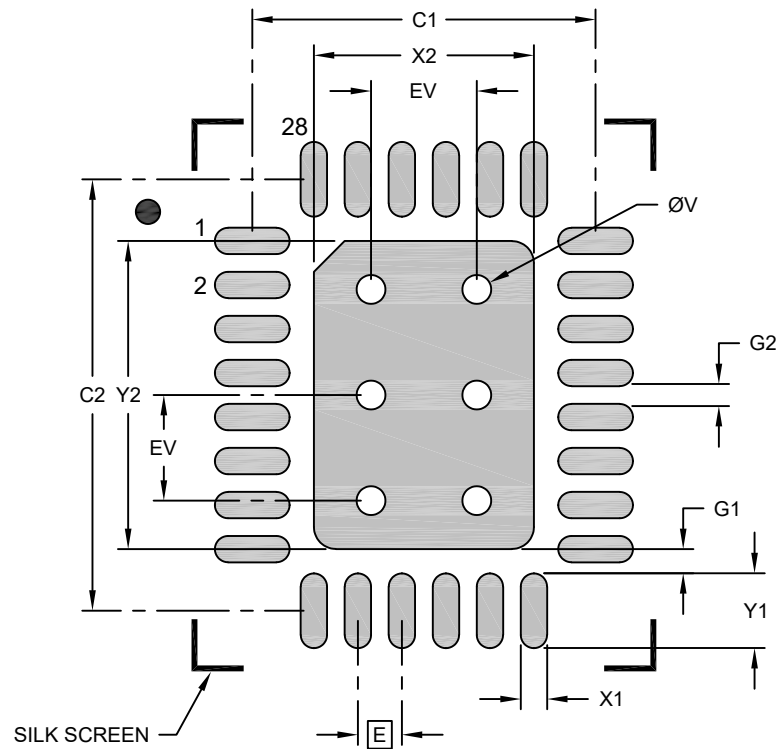
|                                |    | Units | MILLIMETERS |      |       |
|--------------------------------|----|-------|-------------|------|-------|
| Dimension Limits               |    |       | MIN         | NOM  | MAX   |
| Number of Terminals            | N  |       | 28          |      |       |
| Pitch                          | e  |       | 0.50 BSC    |      |       |
| Overall Height                 | A  |       | 0.80        | 0.90 | 1.00  |
| Standoff                       | A1 |       | 0.00        | 0.02 | 0.05  |
| Terminal Thickness             | A3 |       | 0.20 REF    |      |       |
| Overall Length                 | D  |       | 4.00 BSC    |      |       |
| Exposed Pad Length             | D2 |       | 2.30        | 2.40 | 2.50  |
| Overall Width                  | E  |       | 5.00 BSC    |      |       |
| Exposed Pad Width              | E2 |       | 3.30        | 3.40 | 3.50  |
| Terminal Width                 | b  |       | 0.20        | 0.25 | 0.30  |
| Terminal Length                | L  |       | 0.30        | 0.40 | 0.50  |
| Terminal-to-Exposed-Pad        | K  |       | 0.40 REF    |      |       |
| Wettable Flank Step Cut Length | D3 |       | 0.035       | -    | 0.085 |
| Wettable Flank Step Cut Height | A4 |       | 0.10        | -    | 0.19  |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

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**RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |      |      |
| Center Pad Width                 | X2 |             |      | 2.50 |
| Center Pad Length                | Y2 |             |      | 3.50 |
| Contact Pad Spacing              | C1 |             | 3.90 |      |
| Contact Pad Spacing              | C2 |             | 4.90 |      |
| Contact Pad Width (X28)          | X1 |             |      | 0.30 |
| Contact Pad Length (X28)         | Y1 |             |      | 0.85 |
| Contact Pad to Center Pad (X28)  | G1 | 0.28        |      |      |
| Contact Pad to Contact Pad (X24) | G2 | 0.20        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 1.20 |      |

**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23564 Rev B